

User Manual

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1. Introduction

Advanced Lake Biogeochemistry Model (ALBM) is a one-dimensional process-based lake biogeochemistry model that was developed by Dr. Zeli Tan (tanzeli1982@gmail.com) to predict the response of lake thermal and carbon dynamics and water quality to environmental changes (Tan et al., 2015, 2017, 2018). It consists of several modules including those for the radiative transfer, the water/sediment thermal circulation, the water/sediment biogeochemistry, phytoplankton biomass, and the gas diffusive and ebullition transportation. Although the model was originally developed for Arctic lakes (Tan et al., 2015), it has since been used for lakes in diverse environments (Guseva et al., 2020; Guo et al., 2020, 2021). ALBM is a member model of the Inter-Sectoral Impact Model Intercomparison Project (ISIMIP) lake sector (<https://www.isimip.org/impactmodels/details/232/>).

To make this user manual more accessible we will briefly describe the contents of each chapter. [Chapter 2 “Scientific description”](#) describes the physical and biogeochemical processes modeled in ALBM. [Chapter 3 “Model setup”](#) provides the list of model settings and the steps to compile the model code in a Unix-like computer (such as Linux or Mac). [Chapter 4 “Inputs and outputs”](#) describes the input data for model simulations and the output files and variables model simulations produce.

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2. Scientific description

The major physical and biogeochemical processes that ALBM represents are shown in Figure 1. With the current structure, ALBM can represent different lake physical and biogeochemical states and fluxes: (1) thermal regime related (water temperature, sediment temperature, ice thickness, snow thickness, sensible heat flux, latent heat flux, upward shortwave radiation, upward longwave radiation and water-sediment heat conduction), (2) biogeochemistry related, such as dissolved organic carbon (DOC), nitrogen (N₂), oxygen (O₂), carbon dioxide (CO₂), methane (CH₄) and phosphorus (P), gaseous N₂, O₂, CO₂ and CH₄, CO₂ diffusion, CH₄ diffusion and ebullition), and (3) ecosystem related (phytoplankton biomass and chlorophyll).

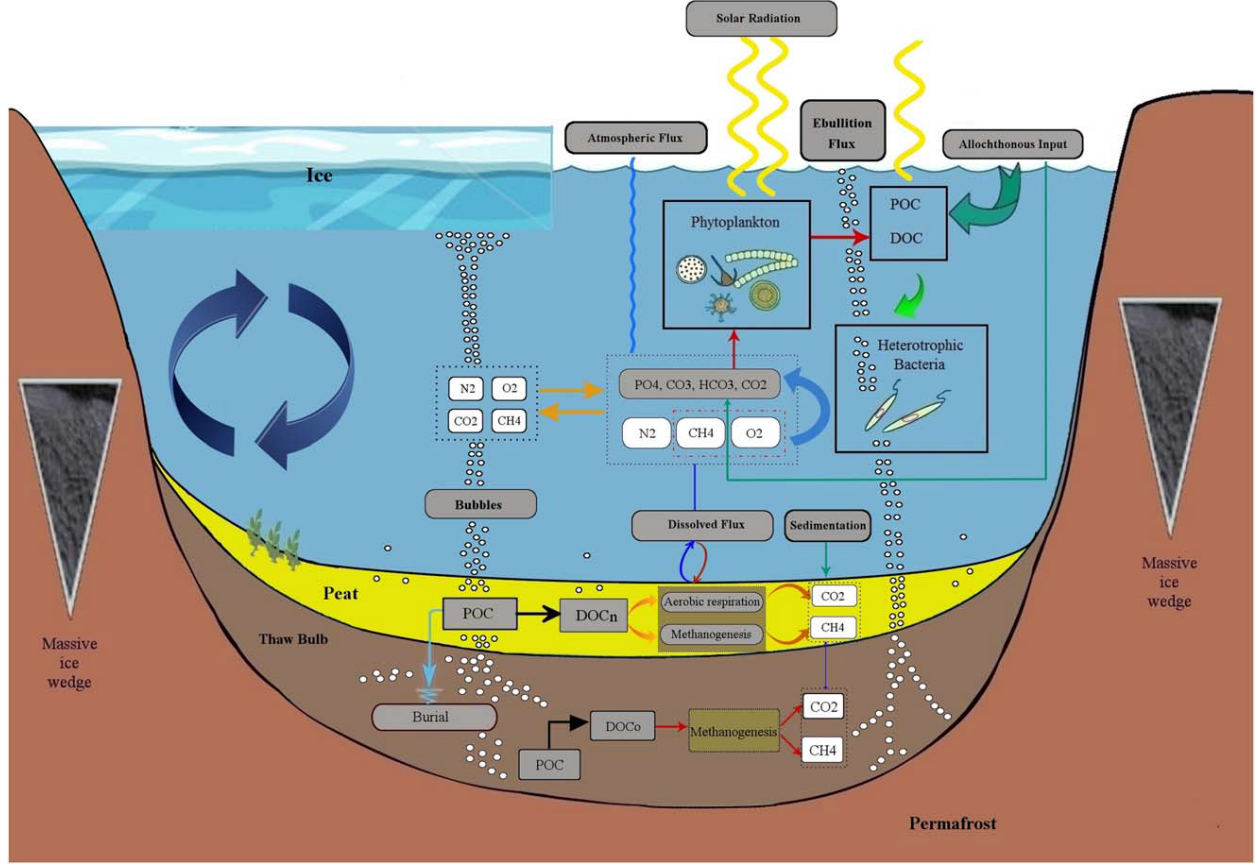


Figure 1. Simplified schematic of thermal, carbon and nutrient dynamics in ALBM.

2.1. Lake thermal dynamics

2.2. Radiative transfer

2.3. Lake carbon dynamics

Our description of the E3SM-LakeBGC model is centered at lake CH_4 dynamics. In the sediment column, the CH_4 dynamics is mainly governed by methanogenesis, ebullition and the interaction of CH_4 with three other major gases, including nitrogen (N_2), oxygen (O_2) and carbon dioxide (CO_2) (Tan et al., 2015):

$$\frac{\partial[\text{CH}_4]}{\partial t} = \frac{\partial}{\partial z} \left(D_{\text{CH}_4}^s \frac{\partial[\text{CH}_4]}{\partial z} \right) + P_{\text{CH}_4} - \alpha_{\text{CH}_4} E, \quad (1)$$

$$\frac{\partial[CO_2]}{\partial t} = \frac{\partial}{\partial z} \left(D_{CO_2}^s \frac{\partial[CO_2]}{\partial z} \right) + P_{CO_2} + \lambda_{CO_2:CH_4} P_{CH_4}, \quad (2)$$

$$\frac{\partial[O_2]}{\partial t} = \frac{\partial}{\partial z} \left(D_{O_2}^s \frac{\partial[O_2]}{\partial z} \right) - \lambda_{O_2:CO_2} P_{CO_2}, \quad (3)$$

$$\frac{\partial[N_2]}{\partial t} = \frac{\partial}{\partial z} \left(D_{N_2}^s \frac{\partial[N_2]}{\partial z} \right) - \alpha_{N_2} E, \quad (4)$$

where $[CH_4]$, $[CO_2]$, $[O_2]$ and $[N_2]$ are the dissolved gas concentrations (mol m^{-3}), $D_{CH_4}^s$, $D_{CO_2}^s$, $D_{O_2}^s$ and $D_{N_2}^s$ are the gas transfer diffusivity in sediment ($\text{m}^2 \text{s}^{-1}$), P_{CH_4} and P_{CO_2} are the CH_4 and CO_2 production rates ($\text{mol m}^{-3} \text{s}^{-1}$) by methanogenesis and aerobic respiration, respectively, E is the gas ebullition rate ($\text{mol m}^{-3} \text{s}^{-1}$), α_{CH_4} and α_{N_2} are the mole fraction of CH_4 and N_2 in formed bubbles, respectively, $\lambda_{CO_2:CH_4}$ is the CO_2 to CH_4 production quotient in methanogenesis, and $\lambda_{O_2:CO_2}$ is the O_2 consumption to CO_2 production quotient in aerobic respiration ($\lambda_{O_2:CO_2} = 1$). $\lambda_{CO_2:CH_4}$ is set to 1 and 3 for active and passive sediment OC, respectively, assuming their different methanogenesis pathways (acetoclastic vs. hydrogenotrophic methanogenesis) (Le Mer & Roger, 2001). As an unreactive gas, N_2 is included due to the important effect of N_2 stripping on ebullition (Tan et al., 2015; Walter et al., 2008).

As described in Tan et al. (2015), P_{CH_4} is calculated as a function of sediment OC content $[OC]$ (mol m^{-3}) and sediment temperature T_s ($^{\circ}\text{C}$):

$$P_{CH_4} = V_{CH_4} \frac{1}{1+\eta_{O_2}[O_2]} f(pH) PQ_{10}^{(T_s-T_{pr})/10} [OC], \quad (5)$$

where V_{CH_4} is the methanogenic rate (s^{-1}) at the reference temperature T_{pr} and anoxic condition,

$\frac{1}{1+\eta_{O_2}[O_2]}$ is the O_2 suppression factor of methanogenesis with the coefficient η_{O_2} of $400 \text{ m}^3 \text{mol}^{-1}$

(Tang et al., 2010), $f(pH) = 10^{-0.2235 \times pH^2 + 2.7727 \times pH - 8.6}$ is the pH effect factor of methanogenesis (Riley et al., 2011), and PQ_{10} is the Q_{10} factor by which the CH_4 production rate increases with a 10°C rise in temperature. Because active and passive OC are different in reactivity,

the parameters of V_{CH_4} and PQ_{10} are set specifically for these two OC pools (Table 1). Likewise, T_{pr} of active and passive sediment OC is set to 0°C and 3.5°C, respectively (Walter & Heimann, 2000; Zhuang et al., 2004). We further assume that CH₄ production would completely shut down when sediment freezes (Walter & Heimann, 2000). The aerobic respiration of sediment OC is calculated as (Tan et al., 2017):

$$P_{CO_2} = V_{CO_2} \frac{[O_2]}{k_{O_2}^{ar} + [O_2]} [OC] \theta_r^{T_s - 20}, \quad (6)$$

where V_{CO_2} is the aerobic respiration rate (s⁻¹) at the reference temperature and O₂-rich condition, $k_{O_2}^{ar}$ is the half-saturation of O₂ for aerobic respiration (mol m⁻³), and θ_r is the temperature-dependence coefficient. In the model, $k_{O_2}^{ar}$ and θ_r are set to 0.047 mol m⁻³ and 1.073, respectively (Hanson et al., 2011). Like V_{CH_4} , V_{CO_2} is different for active and passive sediment OC (Table 1). Because CO₂ is very soluble and O₂ is limited in sediment, we assume that bubbles in sediment only consist of CH₄ and N₂ and the sum of partial pressure of these two gases equals to the hydrostatic pressure exerted on bubbles (Stepanenko et al., 2011). As such, the critical concentration of CH₄ ebullition is calculated as:

$$[CH_4]_{cr} = \Pi H_{CH_4} (P_a + \rho_w g h - [N_2]/H_{N_2}), \quad (7)$$

where Π is the sediment porosity, H_{CH_4} and H_{N_2} are the temperature-dependent henry solubility constants (mol m⁻³ Pa⁻¹) of CH₄ and N₂, respectively, P_a is air surface pressure (Pa), and $\rho_w g h$ is the hydrostatic pressure exerted by water (Pa) that is the product of water density ρ_w , gravitational acceleration g and water depth h . Although theoretically ebullition should start only when [CH₄] exceeds [CH₄]_{cr}, previous field studies showed that it could be initiated when [CH₄] is well below saturation levels (Baird et al., 2004). As such, the ebullition rate E is defined as (Stepanenko et al., 2011):

$$E = \max\{0, c_e([CH_4] - a_e[CH_4]_{cr})\}, \quad (8)$$

by assuming that CH₄ bubbling is activated when [CH₄] reaches to a fraction (a_e) of the saturation level, where c_e is the velocity of bubble formation (s⁻¹). For simplicity, CH₄ bubbles released from sediment are assumed to have diameters in the range of 5–20 mm with a uniform distribution of surface tension (Tan et al., 2015).

In the water column, the dynamics of dissolved CH₄ is governed by diffusion, bubble dissolution and oxidation (Tan et al., 2015, 2017):

$$\frac{\partial[CH_4]}{\partial t} = \frac{\partial}{\partial z} \left(D_{CH_4}^w \frac{\partial[CH_4]}{\partial z} \right) - O_{CH_4} + L_{CH_4}^B, \quad (9)$$

$$\frac{\partial[CO_2]}{\partial t} = \frac{\partial}{\partial z} \left(D_{CO_2}^w \frac{\partial[CO_2]}{\partial z} \right) + O_{CH_4} + f_{AR}L_{Mb} + HR - P_{O_2} + L_{CO_2}^B, \quad (10)$$

$$\frac{\partial[O_2]}{\partial t} = \frac{\partial}{\partial z} \left(D_{O_2}^w \frac{\partial[O_2]}{\partial z} \right) + \phi_{O_2:CO_2}P_{O_2} - 2 \times O_{CH_4} - f_{AR}L_{Mb} - HR + L_{O_2}^B, \quad (11)$$

$$\frac{\partial[N_2]}{\partial t} = \frac{\partial}{\partial z} \left(D_{N_2}^w \frac{\partial[N_2]}{\partial z} \right) + L_{N_2}^B, \quad (12)$$

where $D_{CH_4}^w$, $D_{CO_2}^w$, $D_{O_2}^w$ and $D_{N_2}^w$ are the gas transfer diffusivity in water (m² s⁻¹), O_{CH_4} is the CH₄ oxidation rate (mol m⁻³ s⁻¹), L_{Mb} is the metabolic loss rate of phytoplankton (mol m⁻³ s⁻¹), f_{AR} is the fraction of metabolic loss through autotrophic respiration, HR is the heterotrophic respiration rate in water (mol m⁻³ s⁻¹), P_{O_2} is the CO₂ fixation rate through photosynthesis (mol m⁻³ s⁻¹), $\phi_{O_2:CO_2}$ is the photosynthetic quotient that defines the ratio of O₂ production to CO₂ fixation in photosynthesis, and $L_{CH_4}^B$, $L_{CO_2}^B$, $L_{O_2}^B$ and $L_{N_2}^B$ are the gain of dissolved gases by the dissolution of rising bubbles (mol m⁻³ s⁻¹). $D_{CH_4}^w$, $D_{CO_2}^w$, $D_{O_2}^w$ and $D_{N_2}^w$ are assumed to be equal to the thermal diffusivity of water (Tan et al., 2015). The $\phi_{O_2:CO_2}$ quotient is set to 1.1 in the model (Kirk, 2011). Although many existing CH₄ models use the Michaelis-Menten kinetics to define aerobic CH₄ oxidation (Riley et al., 2011; Tan et al., 2015; Tang et al., 2010), Thottathil et al. (2019) argued that this scheme cannot predict reasonable CH₄ oxidation in the hypolimnion and epilimnion

simultaneously. Instead, Thottathil et al. (2019) suggested using a non-linear CH₄ oxidation scheme to model CH₄ oxidation. Our own experiment also shows that the Michaelis-Menten kinetics would either underestimate CH₄ oxidation in the hypolimnion or overestimate it in the epilimnion (Text S1). Therefore, our model uses the scheme of Thottathil et al. (2019) to calculate CH₄ oxidation:

$$O_{CH_4} = Q_{CH_4} [CH_4]^{\beta_{CH_4}} \left(e^{-\lambda_{O_2}[O_2]} - e^{-(\lambda_{O_2} + \beta_{O_2})[O_2]} \right) OQ_{10}^{(T_w - T_{or})/10}, \quad (13)$$

where Q_{CH_4} is the CH₄ oxidation potential (mol m⁻³ s⁻¹), OQ_{10} is the Q₁₀ factor by which the CH₄ oxidation rate increases with a 10°C rise in temperature, and the coefficients of β_{CH_4} , λ_{O_2} and β_{O_2} are 0.79, 10 and 180, respectively. The reference temperature T_{or} is set to -5.5°C in the model (Zhuang et al., 2004). Following Zimmermann et al. (2021), we disable the O₂ inhibition effect on CH₄ oxidation during lake overturn when CH₄ is sufficient to support the growth of methanotrophic bacteria. Because the dynamics of dissolved OC is not represented in E3SM-LakeBGC, we follow Fang & Stefan (1994) to set HR as a constant (3.62×10^{-9} mol m⁻³ s⁻¹) for all lakes.

The transport of CH₄ bubbles in the water column is represented by modeling three processes, including buoyant rising, gas exchange with ambient water, and bubble expansion (Liang et al., 2011). At location z , the concentration of gases in the bubbles of radius r is determined by a continuity equation (Liang et al., 2011; Tan et al., 2015):

$$\frac{\partial n_m}{\partial t} = -\frac{\partial w^b n_m}{\partial z} - \frac{\partial}{\partial r} \left(\frac{dr}{dt} n_m \right) + \frac{dn_m}{dt} C_b, \quad (14)$$

where n_m is bubble gas concentration ($m = 1$ for N₂, $m = 2$ for O₂, $m = 3$ for CO₂, and $m = 4$ for CH₄), w^b is the rising velocity of bubbles at radius r , C_b is the number of the bubbles with radius r and location z , $\frac{dr}{dt}$ is the bubble radius change rate, and $\frac{dn_m}{dt}$ is the gas exchange rate of a single

bubble. The equations of C_b , $\frac{dr}{dt}$ and $\frac{dn_m}{dt}$ are detailed in Tan et al. (2015). Based on the gas exchange rate of a single bubble and the number of bubbles, the volumetric gas exchange $L_{CH_4}^B$, $L_{CO_2}^B$, $L_{O_2}^B$ and $L_{N_2}^B$ can be calculated as $\int_r C_b \frac{dn_m}{dt} dr$.

To model phytoplankton photosynthesis and metabolism, E3SM-LakeBGC simplifies diverse phytoplankton species into two size-based phytoplankton functional groups as done in ALBM. The large-size group include phytoplankton species that have high growth rates and sinking velocities, such as diatoms, allowing them to flourish in nutrient and light rich areas but preventing them from dominating in quiescent regions (Rousseaux & Gregg, 2015). In contrast, the small-size group includes phytoplankton species that low growth rates and sinking velocities but can sustain in nutrient-poor and windless regions (Rousseaux & Gregg, 2015). The photosynthetic rate P_{O_2} is defined as a function of the chlorophyll concentration [chla] (g Chla m⁻³):

$$P_{O_2} = V_m^0 f(PAR) f(T_w) f(CO_2) f(P) [chla], \quad (15)$$

where V_m^0 is the maximum chlorophyll-specific photosynthetic rate (mol C (g Chla)⁻¹ s⁻¹) under nutrient and light rich conditions, $f(PAR) = (1 - e^{-\alpha PAR/V_m^0}) e^{-\beta PAR/V_m^0}$ is the response function of photosynthesis to photosynthetically active radiation (PAR) in which PAR is in units of mol photons m⁻² s⁻¹, α is the initial slope of photosynthesis-irradiance curve, and β is the photoinhibition factor, $f(T_w) = \theta_p^{T_w-20} - \theta_p^{k_T(T_w-a_T)} + b_T$ is the response function of photosynthesis to temperature in which θ_p is the temperature coefficient ($\theta_p = 1.08$), and k_T , a_T and b_T are fitted coefficients (Hipsey & Hamilton, 2008), $f(CO_2) = [CO_2]/(k_{CO_2} + [CO_2])$ is the response function of photosynthesis to dissolved CO₂ in which k_{CO_2} is the half-saturation constant ($k_{CO_2} = 0.062$ mol m⁻³), and $f(P) = [TP]/(k_{TP} + [TP])$ is the response function of photosynthesis to total

P (TP; mol m⁻³) in which k_{TP} is the half-saturation constant (mol m⁻³). The parameters α , β , V_m^0 and k_{TP} are different for small- and large-size phytoplankton because of their distinct biological traits (Li et al., 2010). Similarly, the coefficients k_T , a_T and b_T are fitted for the optimal, standard and maximum temperature of phytoplankton growth that are different between small- and large-size groups (Hipsey & Hamilton, 2008). In the model, [chla] is simulated proportionally to phytoplankton biomass B_{phyto} (mol C m⁻³) as a ratio that varies in the water column (Wang et al., 2009):

$$[chla] = \frac{B_{phyto}}{(C:Chla)_{surf} - ((C:Chla)_{surf} - (C:Chla)_{min}) \times (\ln(PAR_{surf}) - \ln(PAR)) / 4.605}, \quad (16)$$

where $(C:Chla)_{surf}$ is the surface biomass to chlorophyll ratio (C:Chla), $(C:Chla)_{min}$ is the minimum C:Chla ratio, and PAR_{surf} is the surface PAR level. We assume that the C:Chla ratio must be within the range of $(C:Chla)_{min}$ and $(C:Chla)_{max}$ (Wang et al., 2009). The surface C:Chla is defined as (Wang et al., 2009):

$$(C:Chla)_{surf} = (C:Chla)_{max} - k_{chl} u_0 f(T_w) f(P), \quad (17)$$

where u_0 is the maximum growth rate of phytoplankton at 0°C (d⁻¹) and k_{chl} is the slope of the C:Chla ratio vs. phytoplankton growth rate (gC (g Chla)⁻¹ d), given by Wang et al. (2009). The metabolic loss rate of phytoplankton L_{Mb} is modeled as a function of biomass and temperature (Hanson et al., 2004):

$$L_{Mb} = V_l \theta_l^{T_w - 20} B_{phyto}, \quad (18)$$

where V_l is the phytoplankton-specific metabolic loss potential (s⁻¹) and θ_l is the temperature-dependence coefficient. The coefficient θ_l is set to 1.073 for both small- and large-size phytoplankton (Hanson et al., 2011). The fraction of autotrophic respiration in metabolic loss f_{AR} are set to 0.8 and 0.5 for small- and large-size phytoplankton, respectively (Hanson et al., 2011).

Accordingly, the dynamics of phytoplankton biomass in the water column is defined as (Tan et al., 2017):

$$\frac{\partial B_{phyto}}{\partial t} = V_{settl} \frac{\partial B_{phyto}}{\partial z} + P_{O_2} - L_{Mb}, \quad (19)$$

where V_{settl} is phytoplankton settling velocity (m s^{-1}). V_{settl} is calculated based on Stoke's law as a function of particle diameter, particle density, water density and water viscosity (Hanson et al., 2011). We assume that small phytoplankton can float in the mixing water for better light and nutrient conditions by the buoyant control process (Kromkamp & Walsby, 1990) and a fraction of phytoplankton (f_{resusp}) that falls over the sediment can be resuspended (Saloranta & Andersen, 2007). Accordingly, the total amount of autochthonous OC deposited in lake sediment is calculated as the sum of dead phytoplankton biomass $(1 - f_{AR})L_{Mb}$ and phytoplankton net deposition $(1 - f_{resusp})(V_{settl}B_{phyto})_{bottom}$. For allochthonous OC, its deposition in lake sediment is modeled by the ELM soil erosion (ELM-Erosion) model (Tan et al., 2020; Tan et al., 2021) based on the annual mean water inflow. Following ALBM, intense CH_4 emissions from yedoma lakes are assumed to be fueled by the release of Pleistocene-aged biologically active OC from thawing yedoma permafrost, which increases active OC in sediment (Tan et al., 2015; Walter et al., 2008).

To test the effect of OMP on modeled CH_4 dynamics, we introduce two alternative parameterization schemes for CH_4 production and oxidation. We assume that OMP is promoted in P-limited lakes, such as oligotrophic lakes, due to P acquisition for photosynthesis (Günthel et al., 2020; Yao et al., 2016). Based on the [exponential relationship of OMP to GPP by Bogard et al. \(2014\)](#) and the [negative relationship of OMP to phosphorus levels by Günthel et al. \(2020\)](#), we define [OMP](#) as:

$$OMP = r_{omp}(1 - e^{-a_{omp}P_{O_2}})(1 - b_{omp}[TP]), \quad (20)$$

where r_{omp} is a photosynthesis-driven OMP potential coefficient ($\text{mol CH}_4 (\text{mol C})^{-1}$), and a_{omp} and

b_{omp} are the coefficients for the control of GPP and TP on OMP, respectively. [According to Günthel et al. \(2020\), OMP may be more associated with large phytoplankton cells, colonies and particles.](#)

3. Model setup

3.1. Code structure

The model consists of 26 Fortran code files that correspond to run control, I/O, utility functions, physical processes and biogeochemical processes. Table 1 lists these f90 files and their brief descriptions.

Table 1. Brief description of ALBM code files.

File name	Description
bayesian_mod.f90	subroutines for Monte Carlo based model calibration
bg_const_mod.f90	model constants for biogeochemical processes
bg_utilities_mod.f90	biogeochemical utility subroutines
bLake.f90	main file of the model
boundary_mod.f90	subroutines to process instant boundary conditions
bubble_mod.f90	subroutines to represent the transport of CH ₄ bubbles
carbon_cycle_mod.f90	subroutines for carbon dynamics in the water column
costfunc_mod.f90	cost functions for model calibration
data_buffer_mod.f90	allocated memories for model state and other large variables and subroutines to initiate boundary condition variables
diagenesis_mod.f90	subroutines for carbon dynamics in the sediment column
io_utilities_mod.f90	utility subroutines for I/O
math_utilities_mod.f90	utility subroutines for mathematical algorithms
phy_const_mod.f90	model constants for physical processes
phy_utilities_mod.f90	physics utility subroutines
radiation_io_mod.f90	I/O subroutines for the radiative transfer module
radiation_mod.f90	the radiative transfer module
read_data_mod.f90	I/O subroutines
sensitivity_mod.f90	subroutines for parameter sensitivity analysis

shr_ctrl_mod.f90	variables for model settings
shr_kind_mod.f90	variable precision definitions
shr_param_mod.f90	model calibration parameters
shr_typedef_mod.f90	user-defined variable types
sim_coupler_mod.f90	coupler for different modules and run control subroutines
simulation_mod.f90	regular parallel simulation subroutine
soil_thermal_mod.f90	subroutines for sediment thermal dynamics
thermal_mod.f90	subroutines for water thermal dynamics and ice phenology

3.2. Model settings

ALBM run is controlled by a namelist file *namelist.bLake* in the source code directory. In the namelist file, model settings are classified into eight groups: **general** (lake configurations), **simulation** (switch flags of lake modules and simulation start and end dates), **resolution** (model spatial resolutions), **bayesian** (settings for model calibration and sensitivity analysis), **radiation** (settings for the radiative transfer module), **rundata** (input data), **archive** (module outputs), and **dbg** (model debug use). Table 2 lists all the model settings and their brief descriptions.

Table 2. Brief description of the ALBM settings.

Setting name	Description
general	
run_mode	“regular”, “bayesian” or “sensitivity”
lake_file	the lake file
lakeid_file	the file mapping lake id to lake name
lake_range	selection of lake ids from the lake file for simulation
bthmtry_dir	the directory of lake bathymetry files
param_dir	the directory of lake-specific model parameter files
simulation	
Thermal_Module	switch flag for the thermal dynamics modules
Bubble_Module	switch flag for the CH ₄ bubble module
Diagenesis_Module	switch flag for the sediment carbon dynamics module
Carbon_Module	switch flag for the water carbon dynamics module

Hydro_Module	switch flag for the lake hydrology module
Start_Year	simulation starting year
Start_Month	simulation starting month
Start_Day	simulation starting day
End_Year	simulation ending year
End_Month	simulation ending month
End_Day	simulation ending day
Spinup_Month	spin-up starting month
Spinup_Day	spin-up starting day
nSpinup	the spin-up period in years
resolution	
NWLAYER	the number of water layers
NSLAYER	the number of sediment layers
NRLAYER	the number of bubble radius bands
bayesian	
NMAXSAMPLE	the number of MC parameter samples
sample_range	selection of samples for model calibration or sensitivity simulations
obs_dir	the directory of lake observation files
obs_var	the observational variables for calibration
obs_weight	the weight of observations in the cost function
mc_file	the parameter sample file
sa_file	the cost function output file
radiation	
solar_dir	the directory of solar radiation parameter files
gas_dir	the directory of atmospheric gas optical characteristics files
albedo_dir	the directory of surface albedo characteristics file
co2_file	the atmospheric CO ₂ concentration dynamics file
o3_file	the atmospheric O ₃ concentration dynamics file
aod_file	the atmospheric optical depth (AOD) dynamics file

rundata

forcing_tstep	the time step of atmospheric forcing data (“day” or “hour”)
forcing_dir	the directory of atmospheric forcing data files
hydro_dir	the directory of hydrological forcing data files
tref_file	the reference air temperature file
soc_file	the soil organic carbon file
veg_file	the tree coverage map file
wlnd_file	the wetland distribution map file

archive

archive_tstep	the time step of model outputs (“day” or “hour”)
archive_dir	the directory to save model outputs

dbg

DEBUG	switch flag to dump debug messages (for senior users only)
RESUBMIT	switch flag to resubmit the model simulation (for senior users only)

3.3. Build

Building the ALBM executable consists of three steps: 1) the installation or loading of requisite software, 2) the compilation of dependent packages, and 3) the compilation of the ALBM Fortran code.

First, ALBM needs a Fortran compiler (such as Intel Fortran or GNU Fortran) and a Message Passing Interface (MPI) package (such as OpenMPI or MVAPICH2) for compilation. On Linux clusters, the Fortran compiler and the MPI package are usually already installed. In such cases, they can be loaded by “*module load intel/19.0.3 mvapich2/2.3.1*” or “*module load gcc/4.8.5 mvapich2/2.3.1*”. On Linux or Mac PCs, the easiest way is to install the above-mentioned packages through a package management tool, such as Homebrew for Mac and RPM for RedHat Linux.

Second, ALBM needs a PnetCDF package (or parallel-netcdf before version 1.11.0) to handle the read and write of NetCDF files in the parallel mode. The users can download the source code of the latest PnetCDF package from <https://parallel-netcdf.github.io/wiki/Download.html>. The PnetCDF compilation includes five steps: 1) load the Fortran compiler and the MPI package (e.g., *module load intel/19.0.3 mvapich2/2.3.1*), 2) uncompress the downloaded source code and

switch into the uncompressed source directory, 3) create an installation directory and check the pre-requisite environments for compilation (run *./configure --prefix=install_directory* in source directory), 4) compile the code (run *make* in source directory), and 5) install the package (run *make install* in source directory).

The final step is to compile the ALBM Fortran code. It includes two sub-steps. First, the user needs to modify the make script *make.sh* for an Intel compiler or *gnu_make.sh* for a GNU compiler in the ALBM source directory by (a) pointing NETCDF_HOME to the PnetCDF installation directory and (b) loading the Fortran compiler and MPI package. The second sub-step is to run the command “*./make.sh*” or “*./gnu_make.sh*”. The generated executable *ALBM.exe* will be in the source directory.

3.4. Run

To run ALBM on a Linux cluster, there are two steps: 1) modify the NETCDF_HOME variable and load Fortran compiler and the MPI package (same as the modifications to *make.sh* or *gnu_make.sh*) in the job submission script (*bLakeJob.sub*), and 2) submit a job by “*sbatch bLakeJob.sub*”.

To run ALBM on a Linux or Mac PC, the user should (1) set NETCDF_HOME and LD_LIBRARY_PATH directly in the command terminal and 2) run the model by “*mpirun -np #cores ./ALBM.exe namelist.bLake*”.

4. Inputs and outputs

4.1. Inputs

The atmospheric forcing data ALBM uses must be in a NetCDF-3 file. An example of the NetCDF-3 file structure is shown in Table 3. This file includes ten atmospheric forcing variables.

Table 3. The structure of the atmospheric forcing data file.

<pre>netcdf forcing_obs_Allequash { dimensions: time = UNLIMITED ; // (13880 currently) variables: int date ; date:long_name = "Date of the first data record" ;</pre>
--

```

        date:units = "YYYYMMDD" ;

float tas(time) ;
    tas:long_name = "near-surface daily mean air temperature" ;
    tas:units = "K" ;
    tas:_FillValue = 1.e+20f ;

float tasmin(time) ;
    tasmin:long_name = "near-surface daily minimum air temperature" ;
    tasmin:units = "K" ;
    tasmin:_FillValue = 1.e+20f ;

float tasmax(time) ;
    tasmax:long_name = "near-surface daily maximum air temperature" ;
    tasmax:units = "K" ;
    tasmax:_FillValue = 1.e+20f ;

float hurs(time) ;
    hurs:long_name = "relative humidity" ;
    hurs:units = "%" ;
    hurs:_FillValue = 1.e+20f ;

float ps(time) ;
    ps:long_name = "surface air pressure" ;
    ps:units = "Pa" ;
    ps:_FillValue = 1.e+20f ;

float pr(time) ;
    pr:long_name = "total precipitation" ;
    pr:units = "kg/m2/s" ;
    pr:_FillValue = 1.e+20f ;

float prsn(time) ;
    prsn:long_name = "snowfall flux" ;
    prsn:units = "kg/m2/s" ;
    prsn:_FillValue = 1.e+20f ;

float rsds(time) ;
    rsds:long_name = "surface downwelling shortwave radiation" ;
    rsds:units = "W/m2" ;
    rsds:_FillValue = 1.e+20f ;

float rlds(time) ;
    rlds:long_name = "surface downwelling longwave radiation" ;
    rlds:units = "W/m2" ;
    rlds:_FillValue = 1.e+20f ;

float sfcWind(time) ;
    sfcWind:long_name = "near surface wind speed" ;

```



```

sfcWind:units = "m/s" ;
sfcWind:_FillValue = 1.e+20f ;

// global attributes:

:source = "Arctic Lake Biogeochemistry Model v2.0" ;
:comment = "Observed meteo forcing data for Allequash_Lake Lake from 1/1/1979 to 12/31/2016" ;
}

```

The hydrology input file must also be a NetCDF-3 file. An example of the hydrology input file structure is shown in Table 4. The file includes eight hydrological variables.

Table 4. The structure of the hydrological input data file.

```

netcdf forcing_Harp_ice_CanESM2_hydro_rcp26 {
dimensions:
    time = UNLIMITED ; // (44409 currently)
variables:
    int time(time) ;
        time:long_name = "day" ;
        time:units = "days since 6/1/1978" ;
    float Qsi(time) ;
        Qsi:long_name = "inlet discharge" ;
        Qsi:units = "m3/s" ;
        Qsi:_FillValue = -999.f ;
    float tQsi(time) ;
        tQsi:long_name = "inlet flow temperature" ;
        tQsi:units = "celsius" ;
        tQsi:_FillValue = -999.f ;
    float dQsi(time) ;
        dQsi:long_name = "inlet flow density" ;
        dQsi:units = "kg/m3" ;
        dQsi:_FillValue = -999.f ;
    float DICQsi(time) ;
        DICQsi:long_name = "inlet flow DIC concentration" ;
        DICQsi:units = "uM" ;
        DICQsi:_FillValue = -999.f ;
    float DOCQsi(time) ;
        DOCQsi:long_name = "inlet flow DOC concentration" ;
        DOCQsi:units = "uM" ;
        DOCQsi:_FillValue = -999.f ;
}

```

```

float POCQsi(time) ;
    POCQsi:long_name = "inlet flow POC concentration" ;
    POCQsi:units = "uM" ;
    POCQsi:_FillValue = -999.f ;

float SRPQsi(time) ;
    SRPQsi:long_name = "inlet flow soluble reactive phosphorus" ;
    SRPQsi:units = "uM" ;
    SRPQsi:_FillValue = -999.f ;

float Qso(time) ;
    Qso:long_name = "outlet discharge" ;
    Qso:units = "m3/s" ;
    Qso:_FillValue = -999.f ;

// global attributes:
    :history = "2017-9-18_1:12:8 UTC-05:00 by Zeli Tan" ;
    :comment = "Meteo forcing data for Harp Lake (Ontario, Canada), 6/1/1978 to 12/31/2099" ;
}

```

As described in [Section 3.2](#), there are also other input data needed for model simulations which are available at <https://doi.org/10.6084/m9.figshare.22635064>. If the user needs support for these input data, please contact tanzeli1982@gmail.com.

4.2. Outputs

The ALBM simulation will produce several output NetCDF-3 files with the name in the format of ***bLakeOut.varname.date0_date1.nc*** in which varname is the name of an output variable (such as “watertemp”), date0 is the simulation starting date constructed from the settings of Start_Year, Start_Month and Start_Day (such as “19790101”), and date1 is the simulation ending date constructed from the settings of End_Year, End_Month and End_Day (such as “20060101”). There is a special output file ***bLakeOut.zw.nc*** that provides the information of lake water layers. Table 5 lists the variables that ALBM can produce and their brief description.

Table 5. Brief description of ALBM output variables.

Variable name	Description
thermal dynamics	
icethick	ice cover thickness (meter)
lakeheatf	downward net heat flux at the water-air interface (W m ⁻²)

latentheatf	latent heat flux at the water-air interface (W m^{-2})
lwup	upward longwave radiation at the water-air interface (W m^{-2})
momf	momentum energy flux at the water-air interface ($\text{kg m}^{-1} \text{s}^{-2}$)
sedheatf	upward heat flux at the water-sediment interface (W m^{-2})
sensheatf	sensible heat flux at the water-air interface (W m^{-2})
snowthick	snow cover thickness (meter)
swup	upward shortwave radiation at the water-air interface (W m^{-2})
watertemp	water temperature (K)

carbon dynamics

do	dissolved oxygen (mole m^{-3})
dch4	dissolved methane (mole m^{-3})
dco2	dissolved carbon dioxide (mole m^{-3})
doc	dissolved organic carbon (mole m^{-3})
srp	soluble reactive phosphorus (mole m^{-3})
dno3	dissolved nitrate (not available yet) (mole m^{-3})
dnh4	dissolved ammonium (not available yet) (mole m^{-3})
fch4d	diffusive methane flux at the water-air interface ($\text{mole m}^{-2} \text{d}^{-1}$)
fch4e	ebullitive methane flux at the water-air interface ($\text{mole m}^{-2} \text{d}^{-1}$)
fco2	total carbon dioxide flux at the water-air interface ($\text{mole m}^{-2} \text{d}^{-1}$)

aquatic ecosystem

chl	chlorophyll concentration (g m^{-3})
phytobio	phytoplankton function group biomass (mole m^{-3})

5. References

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